A Dataset of Separated Circles and Triangles

Each image contains one to four disjoint **filled** shapes. The recipe below fixes the sampling distribution, prevents degenerate triangles, and uses rejection sampling to keep the shapes separated. All lengths are measured in the coordinates of the image domain.

1. Sample the number, types, and sizes

Use the following example parameters; they can be rescaled together for a different domain.

Quantity	**Rule**
Image domain	$D = [-1, 1]^2$
Number of shapes	$N \sim \mathrm{Unif}\{1, 2, 3, 4\}$
Shape type	Independently choose disk or triangle, each with probability $1/2$
Disk radius	$r \sim \mathrm{Unif}[0.10, 0.20]$
Triangle side lengths	$a, b, c \overset{\text{iid}}{\sim} \mathrm{Unif}[0.20, 0.30]$
Triangle orientation	$\theta \sim \mathrm{Unif}[0, 2\pi)$, independently
Minimum separation	$\varepsilon = 0.05$

Here, $\mathrm{Unif}\{1, 2, 3, 4\}$ assigns probability $1/4$ to each count, and iid means independent and identically distributed. A disk is $\{x \in \mathbb{R}^2 : \|x - z\|_2 \le r\}$, including its interior.

2. Construct each triangle explicitly

Sample a, b, c as above and set

$$v_1 = (0, 0), \qquad v_2 = (c, 0), \qquad v_3 = (q, h), \qquad q = \frac{b^2 + c^2 - a^2}{2c}, \quad h = \sqrt{b^2 - q^2}.$$

Then $\|v_2 - v_3\|_2 = a$, $\|v_1 - v_3\|_2 = b$, and $\|v_1 - v_2\|_2 = c$. Center the vertices and rotate them:

$$g = \frac{v_1 + v_2 + v_3}{3}, \qquad w_k = Q_\theta(v_k - g), \qquad Q_\theta = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}.$$

The filled triangle is the convex hull of w_1, w_2, w_3. Its final vertices will be $z + w_k$ after a center z is accepted.

Why these triangles are well behaved. More generally, side lengths in $[L, 1.5L]$ always form a nondegenerate triangle: the longest side is at most $1.5L$, while the other two sum to at least $2L$. By the law of cosines, every interior angle α satisfies the sharp bounds

$$\arccos(7/9) \le \alpha \le \arccos(-1/8), \qquad \text{approximately } 38.94^\circ \le \alpha \le 97.18^\circ.$$

The endpoint configurations are $(L, 1.5L, 1.5L)$, opposite L, and $(1.5L, L, L)$, opposite $1.5L$. No additional triangle rejection is needed. The angle bounds hold for *every* triangle; the angles themselves are not sampled independently or uniformly.

3. Reject placements that are too close

For the filled shapes $S_1, \dots, S_N$, the desired geometric conditions are

$$S_i \subset D, \qquad \operatorname{dist}(S_i, S_j) := \min_{x \in S_i,\, y \in S_j} \|x - y\|_2 \geq \varepsilon \quad \text{for every } i < j.$$

For a simple implementation, enclose each shape in a disk centered at its local origin. Define

$$R_i = \begin{cases} r_i, & \text{if shape } i \text{ is a disk,} \\ \max_{k=1,2,3} \|w_{ik}\|_2, & \text{if shape } i \text{ is a triangle.} \end{cases}$$

Sample all centers $z_i = (z_{i,x}, z_{i,y})$ independently and uniformly in D. Accept the complete placement only if

$$\boxed{|z_{i,x}| + R_i \leq 1, \quad |z_{i,y}| + R_i \leq 1 \quad \text{for all } i}$$

and

$$\boxed{\|z_i - z_j\|_2 \geq R_i + R_j + \varepsilon \quad \text{for all } i < j.}$$

Otherwise, resample *all centers*. The pairwise test guarantees separation because any $x \in S_i$ and $y \in S_j$ satisfy

$$\|x - y\|_2 \geq \|z_i - z_j\|_2 - \|x - z_i\|_2 - \|y - z_j\|_2 \geq \varepsilon.$$

This test is sufficient, but conservative. It can reject valid triangle placements whose enclosing disks are too close or cross the domain boundary. It still produces shapes satisfying every required constraint.

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N = random integer from {1, 2, 3, 4}
shapes = sample N types, sizes, and triangle orientations
R = enclosing radius of each shape

repeat:
    z = sample N centers independently from Uniform([-1, 1]^2)
    inside = all(abs(z[i, x]) + R[i] <= 1 and
                 abs(z[i, y]) + R[i] <= 1 for every i)
    separated = all(norm(z[i] - z[j]) >= R[i] + R[j] + eps
                    for every pair i < j)
until inside and separated

translate each shape by its accepted center
rasterize the actual disks and triangles
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Keep N, the types, sizes, and orientations fixed during rejection. Redrawing them after a failed placement can favor smaller objects or smaller counts. If using a trial cap (e.g., 10,000), report a failure or continue sampling centers for the same shapes; do not silently discard the configuration. For $N = 1$, the pairwise test is automatically true.

4. Rasterize and save

Define the binary image by $f(x) = \sum_{i=1}^{N} \mathbf{1}_{S_i}(x)$. On an $m \times m$ grid, evaluate f at pixel centers

$$x_{pq} = \left(-1 + \frac{2(p + 1/2)}{m},\ -1 + \frac{2(q + 1/2)}{m} \right), \qquad p, q = 0, \dots, m - 1.$$

Use $m = 256$ as a starting choice; the grid spacing is $2/m \approx 0.0078$, compared with $\varepsilon = 0.05$. Mark a pixel as 1 if its center lies in any disk or triangle, and 0 otherwise. Use a point-in-triangle test for triangles, *not* their enclosing disks. The separation guarantee applies to the continuous shapes before rasterization.

Save the image, shape count, types, radii or vertices, centers, and random seed so the sample can be reproduced and checked.